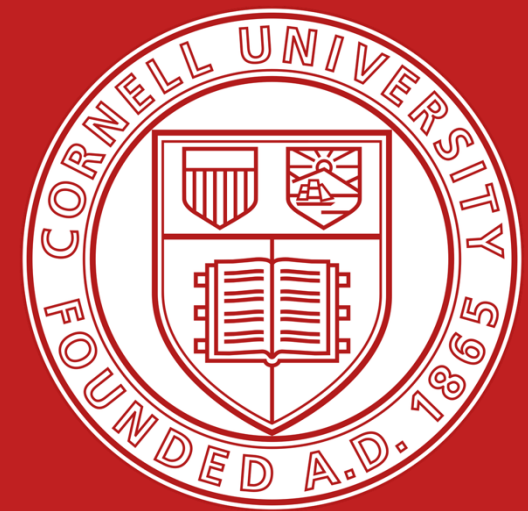




Analysis of a Model Internal Swash Zone: 3D Simulation of Wave Breaking and Flow Structure under Free-Slip and No-Slip Bottom Boundary Conditions

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Background Information

Internal swash zones (ISZs) form where internal waves repeatedly shoal and break along sloping boundaries in stratified lakes and oceans. These breaking events drive turbulence, mixing, and the redistribution of heat, mass, and sediment between the near-bed environment and the stratified interior, with direct consequences for water quality and ecosystem health. A defining but underexplored control on these dynamics is the choice of bottom boundary condition imposed along the slope in ISZ simulations; specifically, whether near-bed friction is fully resolved or neglected entirely.

Motivation

The dynamics of ISZs in lakes such as Cayuga Lake, where large-amplitude internal waves regularly interact with sloping boundaries, remain poorly characterized at high resolution. A key open question is how the bottom boundary condition shapes the breaking process, the no-slip condition resolves the full turbulent boundary layer at significant computational cost, while the free-slip condition avoids near-bed shear entirely at a fraction of that cost. Whether this computational difference translates into fundamentally different flow physics has not been investigated in a fully three-dimensional, turbulence-resolving framework. This gap directly motivates the present study.

Objective

This study examines how free-slip and no-slip bottom boundary conditions alter wave breaking mechanisms, coherent flow structure, and near-bed turbulence in a model ISZ over multiple wave cycles.

Methodology

Simulations were carried out using a high-resolution 3D spectral element solver, modeling a stratified fluid domain representative of a seasonally stratified lake. Finite-amplitude internal waves were generated at the deep-water boundary and allowed to propagate and break over a sloping bed across multiple wave cycles at $Re = 40,000$. Two cases were run — one with a no-slip bottom boundary, which fully resolves friction at the bed, and one with a free-slip bottom boundary, which assumes zero bottom stress. Post-processing was performed in MATLAB, extracting density, vorticity, and turbulent kinetic energy fields from the full 3D simulation snapshots for comparison between cases.

Free-slip

The wall-normal velocity is zero (no penetration), but the tangential velocity has zero normal gradient. The fluid can slide freely along the boundary with no frictional resistance.

No-Slip

All velocity components are zero at the wall. The fluid is brought to rest at the boundary, generating a turbulent boundary layer and wall-normal velocity gradients that produce bottom shear stress.

Results & Analysis

Figures 1a & 1b express the moment the first wave breaks. The free-slip case shows a single, coherent vortex with a smooth rollup. The no-slip case, by contrast, exhibits multiple smaller-scale vortical structures near the slope. The interaction region is more fragmented and turbulent, with finer filaments of alternating vorticity. The no-slip bottom boundary is also visible, and the pycnocline upstream shows more disturbance.

Figures 2a & 2b express the moment the second wave breaks. The free-slip case produces a recognizable vortex rollup, though the upstream pycnocline now shows residual disturbances from the first cycle. In the no-slip case, the interaction region is again more fragmented, multiple small-scale vorticity filaments replace the clean-rollup. The density field in the no-slip case also shows a narrower, more vertically confined upslope.

Figures 3a & 3b express the moment the third wave breaks. The free-slip case shows a vertically extensive interaction zone with multiple vortex cores and significantly more mixed fluid above the pycnocline. The no-slip case is more laterally compact, with vorticity concentrated at the slope. Notably, the no-slip density field shows sharper upslope intrusions but less vertical lofting of mixed fluid, while the pycnocline upstream sits lower, indicating more cumulative erosion from below.

Figures 4a & 4b express an intermediary step between the third and fourth wave break. The free-slip case retains an energetic, vertically tall mixing with fine-scale vorticity billows reaching nearly to the top boundary. The no-slip case shows substantially more diffuse vorticity spread across a wider horizontal extent, but with weaker amplitudes. The no-slip density field is also notably more blended, the upslope density structure appears as a thinner, more continuous layer rather than distinct lofted plumes.

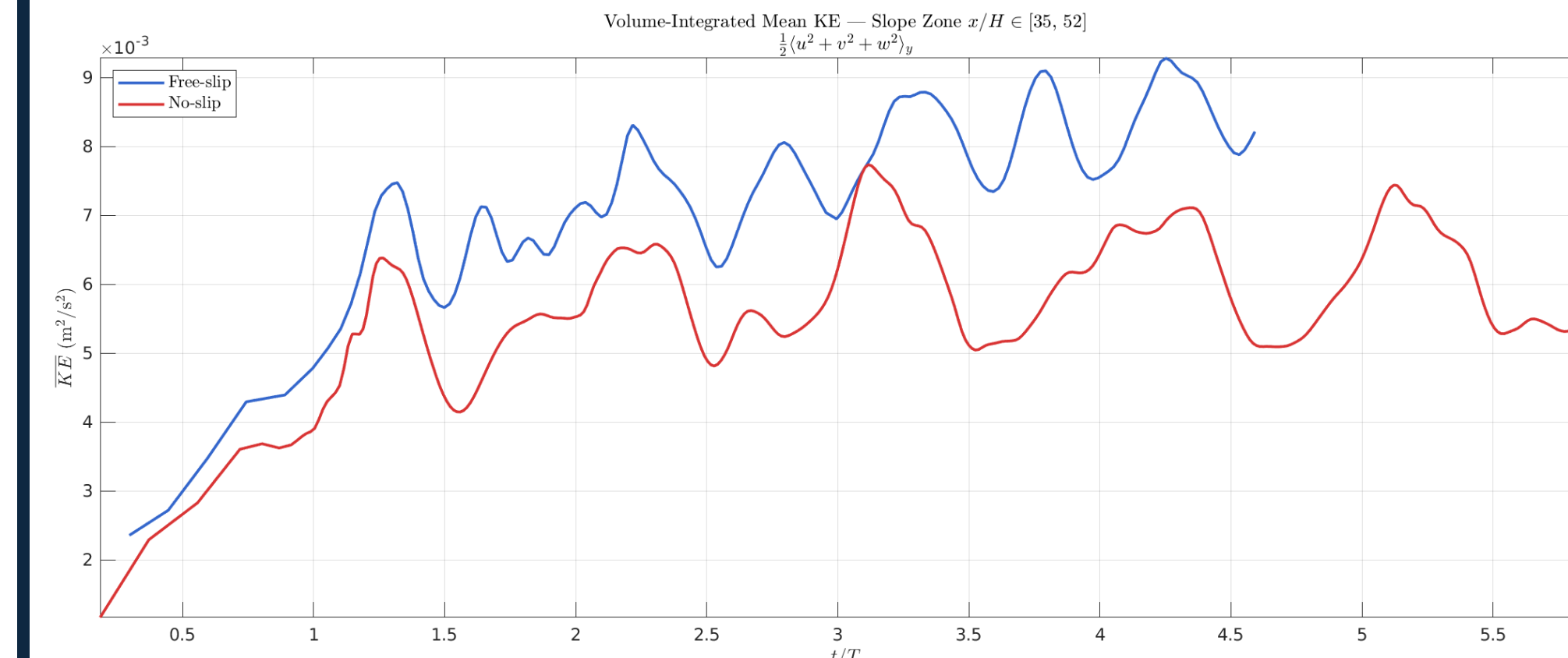


Figure 5. Volume-Integrated Mean Kinetic Energy

Figure 5 represents mean kinetic energy in the slope zone. The free-slip case consistently carries more kinetic energy than no-slip across all wave cycles. Both cases exhibit clear oscillatory peaks corresponding to successive wave arrivals, but the no-slip signal has larger relative fluctuations and deeper troughs between cycles.

Conclusion

The free-slip case produces coherent vortices that loft mixed fluid vertically, no-slip fragments these into smaller-scale, boundary-layer driven turbulence from the first wave impact. Free-slip sustains consistently higher-volume integrated kinetic energy in the slope zone, while no-slip exhibits deeper inter-cycle troughs due to frictional dissipation. Despite lower kinetic energy, the no-slip case drives more cumulative pycnocline erosion, by later cycles the stratification is notably more blended and the pycnocline sits lower. Free-slip concentrates mixing in localized, energetic bursts; no-slip redistributes it into diffuse turbulence that more thoroughly erodes the stratification over successive wave cycles.

Future Work

Future work includes quantifying mass transport and offshore transmission of mixed fluid over multiple wave cycles. Computing bottom shear stresses (no-slip) and near-bed vertical velocities to assess resuspension potential linked to return flows. Lagrangian particle tracking to characterize the potential nepheloid layer development behind the turbulent structures.

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